Continuity of the stabilized bilinear form for the linearized porous Navier–Stokes problem: a self-contained proof for the ASGS method

Abstract

We give a complete and self-contained proof of the continuity estimate for the stabilized (ASGS) bilinear form associated with the linearized porous Navier–Stokes equations, in the form stated as the continuity lemma of the main text. The proof follows, step by step, the proof of Lemma 2 in Codina [1], adapting every estimate to the presence of the porosity field α in the differential operator, in the stabilization parameters and in the working norm. We also record the modifications needed to obtain the corresponding continuity estimate with respect to the interpolation error, from which the convergence theorem of the main text follows by the standard Céa-type argument of [1]. Along the way we state precisely all the hypotheses that the proof requires, and we amend a few minor inaccuracies, both in the main text (the smallness condition on the penalty parameter ε, the definition of the broken norm, and the constants of the inverse estimates applied to α-weighted quantities) and in [1] (a sign in the inter-element term of Eq. (66) and the regularity exponent for the pressure in Eq. (77)); see Section 5.

1 Setting

1.1 The linearized problem and the stabilized bilinear form

Let $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, be a bounded polyhedral domain with boundary Γ. As in the analysis section of the main text, we consider the ASGS method applied to the linearized porous Navier–Stokes problem with homogeneous Dirichlet boundary conditions on the whole of Γ, uniform kinematic viscosity $\nu > 0$ and a uniform, isotropic resistance $\boldsymbol{\sigma} = \sigma\mathbb{I}$, $\sigma \geq 0$. The linearized differential operator is, for $U = [\boldsymbol{u}; p]$,

$$\mathcal{L}_{\boldsymbol{a}} U = \begin{bmatrix} \alpha\, \boldsymbol{a} \cdot \nabla \boldsymbol{u} - 2\nabla \cdot \big(\alpha\nu \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{u}\big) + \alpha \nabla p + \sigma \boldsymbol{u} \\ \varepsilon p + \nabla \cdot (\alpha \boldsymbol{u}) \end{bmatrix}, \tag{1.1}$$

where $\boldsymbol{a}$ is the given advection field, $\alpha : \Omega \to (0, 1]$ is the fluid volume fraction, $\varepsilon \geq 0$ is the penalty parameter, and $\overset{\mathrm{DS}}{\Pi}$ denotes the (pointwise, orthogonal) projection extracting the deviatoric–symmetric part of a second-order tensor, so that $\big|\overset{\mathrm{DS}}{\Pi}\mathbf{T}\big| \leq |\mathbf{T}|$ pointwise for any tensor $\mathbf{T}$. Throughout we use the abbreviations

$$X(V) \coloneqq \boldsymbol{a} \cdot \nabla \boldsymbol{v} + \nabla q, \qquad G(\boldsymbol{v}) \coloneqq 2\nu\, \nabla \cdot \big(\alpha \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{v}\big) - \sigma \boldsymbol{v}, \qquad V = [\boldsymbol{v}; q], \tag{1.2}$$

so that the momentum component of $\mathcal{L}_{\boldsymbol{a}} U$ reads $\alpha X(U) - G(\boldsymbol{u})$. Under the assumption $\nabla \cdot (\alpha \boldsymbol{a}) = 0$ (see Assumption 3 below) and with the simplification of the main text (removal of the term proportional to $\nabla \cdot (\alpha \boldsymbol{a})$, which vanishes anyway in the present setting), the formal adjoint has momentum component $-\alpha X(V) - G(\boldsymbol{v})$ and mass component $\varepsilon q - \nabla \cdot (\alpha \boldsymbol{v})$.

Let $\mathcal{T}_h$ be a finite element partition of $\overline{\Omega}$ of elements K with diameters h_K, and let $\mathcal{X}_{h0} = \mathcal{V}_{h0} \times \mathcal{Q}_{h0}$, with $\mathcal{V}_{h0} \subset H^1_0(\Omega)^d$ and $\mathcal{Q}_{h0} \subset H^1(\Omega) \cap \mathcal{Q}_0$, be the continuous finite element spaces of polynomial degrees $k_u \geq 1$ and $k_p \geq 1$ used for the velocity and the pressure ($\mathcal{Q}_0$ being

$L^2(\Omega)$, or its zero-mean subspace when $\varepsilon = 0$). The $L^2(\omega)$ inner product is written $(\cdot,\cdot)_\omega$, with ω omitted when $\omega = \Omega$, and for elementwise defined quantities we write

$$(f, \tau g)_h \coloneqq \sum_{K\in\mathcal{T}_h} \int_K f\, \tau\, g \,\mathrm{d}\Omega, \qquad \|\tau^{1/2} f\|_h \coloneqq \Big(\sum_{K\in\mathcal{T}_h} \int_K \tau\, |f|^2 \,\mathrm{d}\Omega \Big)^{1/2}, \tag{1.3}$$

where τ may take a different (constant) value τ_K on each element. Note that the broken norm is the square root of the *sum of squares* of the elemental contributions (cf. Section 5).

Restricted to the present setting, the stabilized bilinear form of the main text, $B_{\mathrm{ASGS}}(\boldsymbol{a}; U_h, V_h) = B(\boldsymbol{a}; U_h, V_h) - \sum_K \langle \mathcal{L}^* V_h, \boldsymbol{\tau}_K \mathcal{L} U_h \rangle_K$, reads, for $U_h = [\boldsymbol{u}_h; p_h]$ and $V_h = [\boldsymbol{v}_h; q_h]$ in $\mathcal{X}_{h0}$,

$$\begin{aligned} B_{\mathrm{ASGS}}(\boldsymbol{a}; U_h, V_h) = 2\nu\big(\overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}_h, \alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u}_h\big) + \big(\boldsymbol{v}_h, \alpha X(U_h)\big) + \sigma\big(\boldsymbol{v}_h, \boldsymbol{u}_h\big) + \varepsilon\big(q_h, p_h\big) + \big(q_h, \nabla\cdot(\alpha \boldsymbol{u}_h)\big) \\ + \big(\alpha X(V_h) + G(\boldsymbol{v}_h),\ \tau_1\big[\alpha X(U_h) - G(\boldsymbol{u}_h)\big]\big)_h \\ + \big(\nabla\cdot(\alpha \boldsymbol{v}_h) - \varepsilon q_h,\ \tau_2\big[\varepsilon p_h + \nabla\cdot(\alpha \boldsymbol{u}_h)\big]\big)_h, \end{aligned} \tag{1.4}$$

together with $L_{\mathrm{S}}(\boldsymbol{a}; V_h) = (\boldsymbol{v}_h, \boldsymbol{f}) + \big(\alpha X(V_h) + G(\boldsymbol{v}_h), \tau_1 \boldsymbol{f}\big)_h$. For the exact solution U of the linearized continuous problem (assumed smooth enough, see Section 4) one has $\mathcal{L}_{\boldsymbol{a}} U = F$ in $L^2(K)$ for every K, and therefore the method is consistent:

$$B_{\mathrm{ASGS}}(\boldsymbol{a}; U - U_h, V_h) = 0 \qquad \forall\, V_h \in \mathcal{X}_{h0}. \tag{1.5}$$

1.2 Stabilization parameters and working norm

On each element K the stabilization parameters of the main text are

$$\tau_{1,K} = \frac{1}{\alpha_K\, \tau_{1,\mathrm{NS}}^{-1} + \sigma}, \qquad \tau_{2,K} = \frac{h_K^2}{c_1\, \alpha_K\, \tau_{1,\mathrm{NS}}}, \qquad \tau_{1,\mathrm{NS}}^{-1} = c_1 \frac{\nu}{h_K^2} + c_2 \frac{|\boldsymbol{a}|_{\infty,K}}{h_K}, \tag{1.6}$$

where $c_1, c_2 > 0$ are the algorithmic constants, $|\boldsymbol{a}|_{\infty,K} \coloneqq \sup_K |\boldsymbol{a}|$ (Euclidean norm), and $\alpha_K \coloneqq \alpha_{\infty,K} \coloneqq \sup_K \alpha$; we also write $\alpha_{0,K} \coloneqq \inf_K \alpha > 0$. It is convenient to define, in analogy with [1],

$$\varphi_{1,K} \coloneqq \alpha_K\, \tau_{1,\mathrm{NS}}^{-1} = \alpha_K \Big(c_1 \frac{\nu}{h_K^2} + c_2 \frac{|\boldsymbol{a}|_{\infty,K}}{h_K} \Big), \qquad \text{so that} \qquad \tau_{1,K} = \big(\varphi_{1,K} + \sigma\big)^{-1}, \tag{1.7}$$

and

$$\widetilde{\sigma}_\alpha\big|_K \coloneqq \frac{\tau_{1,\mathrm{NS}}^{-1} \sigma}{\tau_{1,\mathrm{NS}}^{-1} + \sigma/\alpha_K} = \frac{\varphi_{1,K}\, \sigma}{\varphi_{1,K} + \sigma} = \sigma - \sigma^2 \tau_{1,K} = \sigma\, \varphi_{1,K}\, \tau_{1,K} \ \geq 0. \tag{1.8}$$

We will drop the element subscript (τ_1, τ_2, φ_1, α_K, h, $|\boldsymbol{a}|_\infty$) whenever the elementwise character is clear from the context. The working norm here is the ASGS working norm $|||\cdot|||_{\mathrm{A}}$ of the main text; as this note treats only the ASGS method, the subscript is redundant and we drop it, writing simply $|||\cdot|||$. It is

$$|||V_h||| \coloneqq \Big(\nu \big\| \alpha^{1/2} \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{v}_h \big\|^2 + \big\| \widetilde{\sigma}_\alpha^{1/2} \boldsymbol{v}_h \big\|_h^2 + \varepsilon \big\| q_h \big\|^2 + \big\| \tau_1^{1/2} \alpha X(V_h) \big\|_h^2 + \big\| \tau_2^{1/2} \nabla\cdot(\alpha \boldsymbol{v}_h) \big\|_h^2 \Big)^{1/2}. \tag{1.9}$$

1.3 Assumptions

Throughout, C denotes a generic positive constant, possibly different at each occurrence, that may depend on d, on the polynomial degrees, on the shape regularity of the mesh family, on the algorithmic constants c_1, c_2 and on the constants C_α, C_2, c_J, c'_J introduced below, but *not* on the mesh size, nor on $\nu, \sigma, \varepsilon, \boldsymbol{a}$ or α. We assume:

Assumption 1 (Data)**.** $\nu > 0$ and $\sigma \geq 0$ are constants, $\boldsymbol{f} \in L^2(\Omega)^d$, and the penalty parameter satisfies

$$\varepsilon \;\leq\; C_2\, c_1 \min_{K\in\mathcal{T}_h} \frac{\alpha_K^2\, \tau_{1,K}}{h_K^2}, \qquad 0 \leq C_2 < 1. \tag{1.10}$$

Assumption 2 (Porosity)**.** $\alpha \in W^{1,\infty}(\Omega)$ with $0 < \alpha \leq 1$ in $\overline{\Omega}$, and the porosity variations are resolved by the mesh: there is $C_\alpha \geq 0$ such that

$$h_K\, \|\nabla\alpha\|_{L^\infty(K)} \leq C_\alpha\, \alpha_{0,K} \qquad \forall\, K \in \mathcal{T}_h. \tag{1.11}$$

In particular $\alpha_{\infty,K} \leq \delta_\alpha\, \alpha_{0,K}$ with $\delta_\alpha := 1 + C_\alpha$, so that $\alpha_{0,K} \leq \alpha(\boldsymbol{x}) \leq \alpha_K \leq \delta_\alpha\, \alpha_{0,K}$ for $\boldsymbol{x} \in K$.

Assumption 3 (Advection)**.** $\boldsymbol{a} \in C(\overline{\Omega})^d$ (in the application, $\boldsymbol{a}$ is the finite element velocity of the previous iteration) and $\nabla\cdot(\alpha\boldsymbol{a}) = 0$ in the sense of distributions, i.e. $\int_\Omega \alpha\boldsymbol{a}\cdot\nabla\psi\,\mathrm{d}\Omega = 0$ for all $\psi \in W_0^{1,1}(\Omega)$.

Assumption 4 (Mesh)**.** The family of partitions $\{\mathcal{T}_h\}_{h>0}$ is nondegenerate (shape regular and locally quasi-uniform). In particular: (i) the inverse estimates

$$\|\nabla\psi_h\|_K \leq \frac{C_{\mathrm{inv}}}{h_K}\|\psi_h\|_K, \qquad \|\psi_h\|_{L^\infty(K)} \leq \frac{C_{\mathrm{inv}}}{h_K^{d/2}}\|\psi_h\|_K, \tag{1.12}$$

hold for every polynomial ψ_h of the degrees employed (componentwise for vector- and tensor-valued functions); (ii) the diameters of neighboring elements are uniformly comparable, each element has a uniformly bounded number of faces, and $\mathrm{meas}(\Gamma^b) \leq C\, h_K^{d-1}$ for each face Γ^b of each element K.

Assumption 5 (Jump control, needed only when $\sigma > 0$)**.** For every pair of elements K_1, K_2 sharing a face,

$$c_J\, \varphi_{1,K_1} \leq \varphi_{1,K_2} \leq c_J'\, \varphi_{1,K_1}, \qquad 0 < c_J \leq 1 \leq c_J'. \tag{1.13}$$

Remark 1.1. Assumption 5 is the analogue of condition (39) in [1]. By Assumption 2 the factors α_{K_i} of neighboring elements are automatically comparable ($\alpha_{K_1} \leq \delta_\alpha\, \alpha_{K_2}$, since both suprema are bounded below by the value of α on the shared face up to the factor δ_α), and by Assumption 4 so are h_{K_1} and h_{K_2}. Hence (1.13) reduces to the comparability of $|\boldsymbol{a}|_{\infty,K_1}$ and $|\boldsymbol{a}|_{\infty,K_2}$ relative to ν/h_K, a mild requirement on the advection field already implicit in [1].

2 Auxiliary results

Lemma 2.1 (Parameter inequalities)**.** *On each element K (subscripts omitted) there hold:*

- (P1) $\sigma\tau_1 \leq 1$, $\varphi_1\tau_1 \leq 1$, $\tau_1 \leq \varphi_1^{-1}$;
- (P2) $c_1\,\nu\,\alpha_K \leq \varphi_1 h^2$ *and* $c_2\,\alpha_K\,|\boldsymbol{a}|_\infty\, h \leq \varphi_1 h^2$; *in particular* $\nu\,\tau_1\,\alpha_K \leq h^2/c_1$;
- (P3) $\varphi_1 h^2 = c_1\,\alpha_K^2\,\tau_2$; *in particular* $\varphi_1^{1/2} h = c_1^{1/2}\,\alpha_K\,\tau_2^{1/2} \leq c_1^{1/2}\,\tau_2^{1/2}$;
- (P4) $\widetilde{\sigma}_\alpha \leq \min\{\sigma, \varphi_1\}$ *and* $\widetilde{\sigma}_\alpha\,\tau_1 \leq 1$;
- (P5) *under* (1.10), $\varepsilon\tau_2 \leq C_2\,\varphi_1\tau_1 \leq C_2 < 1$; *moreover,* $\varepsilon h^2 \leq C_2\, c_1\,\alpha_K^2\,\tau_1$, *and hence* $\varepsilon^{1/2} \leq c_1^{1/2}\,\alpha_K\,\tau_1^{1/2}/h$.

Proof. (P1) and (P2) are immediate from (1.6)–(1.7) and $\tau_1^{-1} = \varphi_1 + \sigma$. For (P3), $\varphi_1 h^2 = \alpha_K h^2 \tau_{1,\mathrm{NS}}^{-1} = \alpha_K^2\, c_1 \frac{h^2}{c_1 \alpha_K \tau_{1,\mathrm{NS}}} = c_1 \alpha_K^2 \tau_2$, and $\alpha_K \le 1$ by Assumption 2. For (P4), use (1.8): $\widetilde{\sigma}_\alpha = \sigma\varphi_1/(\varphi_1 + \sigma)$ is bounded by both factors of the harmonic-type mean, and $\widetilde{\sigma}_\alpha \tau_1 = \sigma\varphi_1\tau_1^2 \le (\sigma\tau_1)(\varphi_1\tau_1) \le 1$. For (P5): from (1.10) on element K, $\varepsilon \le C_2 c_1 \alpha_K^2 \tau_1/h^2$; multiplying by τ_2 and using (P3), $\varepsilon\tau_2 \le C_2\tau_1\varphi_1 \le C_2$. The remaining statements follow from $\alpha_K \le 1$ and $C_2 < 1$. □

Lemma 2.2 (Weighted inverse estimates)**.** *Let Assumptions 2 and 4 hold. There is a constant $\overline{C}_{\mathrm{inv}} = \overline{C}_{\mathrm{inv}}(C_{\mathrm{inv}}, C_\alpha, d)$ such that, for all finite element functions $\boldsymbol{w}_h \in \mathcal{V}_{h0}$, $r_h \in \mathcal{Q}_{h0}$, and all $K \in \mathcal{T}_h$,*

$$\big\|\nabla \cdot (\alpha \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{w}_h)\big\|_K \le \frac{\overline{C}_{\mathrm{inv}}}{h_K}\, \alpha_K^{1/2}\, \big\|\alpha^{1/2} \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{w}_h\big\|_K, \tag{2.1}$$

$$\big\|\alpha^{1/2} \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{w}_h\big\|_K \le \frac{C_{\mathrm{inv}}}{h_K}\, \alpha_K^{1/2}\, \|\boldsymbol{w}_h\|_K, \tag{2.2}$$

$$\big\|\nabla \cdot (\alpha \boldsymbol{w}_h)\big\|_K \le \frac{\overline{C}_{\mathrm{inv}}}{h_K}\, \alpha_K\, \|\boldsymbol{w}_h\|_K, \tag{2.3}$$

$$\|\alpha\, \boldsymbol{a} \cdot \nabla \boldsymbol{w}_h\|_K \le \frac{C_{\mathrm{inv}}}{h_K}\, \alpha_K\, |\boldsymbol{a}|_{\infty,K}\, \|\boldsymbol{w}_h\|_K, \qquad \|\alpha \nabla r_h\|_K \le \frac{C_{\mathrm{inv}}}{h_K}\, \alpha_K\, \|r_h\|_K. \tag{2.4}$$

Proof. Write $\mathbf{M} := \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{w}_h$, a polynomial tensor field on K (the projection $\overset{\mathrm{DS}}{\Pi}$ acts pointwise with constant coefficients). By the product rule, $\nabla \cdot (\alpha\mathbf{M}) = \alpha\,\nabla \cdot \mathbf{M} + \mathbf{M}\,\nabla\alpha$. For the first contribution, using $\|\nabla \cdot \mathbf{M}\|_K \le \sqrt{d}\,\|\nabla\mathbf{M}\|_K$, the inverse estimate (1.12), and $\|\mathbf{M}\|_K \le \alpha_{0,K}^{-1/2}\|\alpha^{1/2}\mathbf{M}\|_K$,

$$\|\alpha\,\nabla\cdot\mathbf{M}\|_K \le \alpha_{\infty,K}\sqrt{d}\,\frac{C_{\mathrm{inv}}}{h_K}\,\|\mathbf{M}\|_K \le \sqrt{d}\,\frac{C_{\mathrm{inv}}}{h_K}\,\frac{\alpha_{\infty,K}}{\alpha_{0,K}^{1/2}}\,\|\alpha^{1/2}\mathbf{M}\|_K \le \sqrt{d\,\delta_\alpha}\,\frac{C_{\mathrm{inv}}}{h_K}\,\alpha_K^{1/2}\,\|\alpha^{1/2}\mathbf{M}\|_K,$$

since $\alpha_{\infty,K}/\alpha_{0,K}^{1/2} = (\alpha_{\infty,K}/\alpha_{0,K})^{1/2}\alpha_{\infty,K}^{1/2} \le \delta_\alpha^{1/2}\alpha_K^{1/2}$ by Assumption 2. For the second contribution, by (1.11),

$$\|\mathbf{M}\,\nabla\alpha\|_K \le \|\nabla\alpha\|_{L^\infty(K)}\|\mathbf{M}\|_K \le \frac{C_\alpha\,\alpha_{0,K}}{h_K}\,\alpha_{0,K}^{-1/2}\|\alpha^{1/2}\mathbf{M}\|_K \le \frac{C_\alpha}{h_K}\,\alpha_K^{1/2}\,\|\alpha^{1/2}\mathbf{M}\|_K.$$

This proves (2.1) with $\overline{C}_{\mathrm{inv}} := \sqrt{d\,\delta_\alpha}\,C_{\mathrm{inv}} + C_\alpha$. Estimate (2.2) follows from $\alpha \le \alpha_K$ on K, the pointwise contraction property $|\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{w}_h| \le |\nabla\boldsymbol{w}_h|$ and (1.12). For (2.3), expand $\nabla\cdot(\alpha\boldsymbol{w}_h) = \alpha\nabla\cdot\boldsymbol{w}_h + \nabla\alpha\cdot\boldsymbol{w}_h$ and argue as above (using $\alpha_{0,K} \le \alpha_K$). The estimates (2.4) are immediate from $\alpha \le \alpha_K$, $|\boldsymbol{a}| \le |\boldsymbol{a}|_{\infty,K}$ and (1.12). □

Remark 2.3. Lemma 2.2 makes precise the inverse estimates used in the stability proof of the main text for the α-weighted second-order term: since α is *not* piecewise polynomial, the constant is not the bare C_{inv} but the slightly larger $\overline{C}_{\mathrm{inv}}$, which incorporates the porosity-resolution constant C_α of (1.11). All statements of the main text remain valid verbatim upon replacing C_{inv} by $\overline{C}_{\mathrm{inv}}$ in the conditions on c_1 (cf. Section 5).

Lemma 2.4 (Jump of τ_1)**.** *Let Assumption 5 hold and let K_1, K_2 share the face Γ^b. Setting $[\![\tau_1]\!] := \tau_{1,K_1} - \tau_{1,K_2}$, there holds, for $i \in \{1,2\}$,*

$$\sigma\,\big|[\![\tau_1]\!]\big| \le C\,\frac{\sigma\,\varphi_{1,K_i}}{(\varphi_{1,K_i} + \sigma)^2} = C\,\widetilde{\sigma}_\alpha\big|_{K_i}\,\tau_{1,K_i}. \tag{2.5}$$

Proof. We have $[\![\tau_1]\!] = \dfrac{\varphi_{1,K_2} - \varphi_{1,K_1}}{(\varphi_{1,K_1} + \sigma)(\varphi_{1,K_2} + \sigma)}$, and (1.13) gives $|\varphi_{1,K_1} - \varphi_{1,K_2}| \le \max\{c_J' - 1,\, 1 - c_J\}\,\varphi_{1,K_i}$ as well as $\varphi_{1,K_2} + \sigma \ge \min\{c_J, 1\}\,(\varphi_{1,K_1} + \sigma)$ (and symmetrically), from which (2.5) follows with the identity (1.8). □

3 The continuity estimate

Lemma 3.1 (Continuity)**.** *Let Assumptions 1 to 5 hold, with $c_1, c_2 > 0$. Then there exists $C > 0$, independent of the mesh size and of $\nu, \sigma, \varepsilon, \boldsymbol{a}, \alpha$, such that*

$$B_{\mathrm{ASGS}}(\boldsymbol{a}; U_h, V_h) \le C\Big(\big\|\tau_2^{1/2} h^{-1} \boldsymbol{u}_h\big\|_h + \big\|\tau_1^{1/2} h^{-1} p_h\big\|_h\Big)\, |||V_h||| \qquad \forall\, U_h, V_h \in \mathcal{X}_{h0}. \tag{3.1}$$

Moreover, the proof yields the sharper bound

$$B_{\mathrm{ASGS}}(\boldsymbol{a}; U_h, V_h) \le C\Big(\big\|\alpha_K\, \tau_2^{1/2} h^{-1} \boldsymbol{u}_h\big\|_h + \big\|\alpha_K\, \tau_1^{1/2} h^{-1} p_h\big\|_h\Big)\, |||V_h|||, \tag{3.2}$$

where α_K denotes the piecewise constant function equal to $\alpha_{\infty,K}$ on each $K \in \mathcal{T}_h$; (3.1) follows from (3.2) since $\alpha \le 1$.

Proof. Throughout the proof we write $U = U_h = [\boldsymbol{u}; p]$ and $V = V_h = [\boldsymbol{v}; q]$, dropping the subscript h of the unknowns for conciseness, and we omit the element subscript K in $\tau_1, \tau_2, \varphi_1, \alpha_K, h, |\boldsymbol{a}|_\infty$ inside elementwise sums. The proof follows the steps of the proof of Lemma 2 in [1].

Step 0: term decomposition. Expanding the products in (1.4) we obtain $B_{\mathrm{ASGS}}(\boldsymbol{a}; U, V) = \sum_{j=1}^{18} T_j$, with

$$T_1 := 2\nu\big(\overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}, \alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u}\big), \tag{3.3}$$
$$T_2 := \big(\boldsymbol{v}, \alpha X(U)\big), \tag{3.4}$$
$$T_3 := \sigma\,(\boldsymbol{v}, \boldsymbol{u}), \tag{3.5}$$
$$T_4 := \big(q, \nabla\cdot(\alpha \boldsymbol{u})\big), \tag{3.6}$$
$$T_5 := \varepsilon\,(q, p), \tag{3.7}$$
$$T_6 := -4\nu^2\big(\tau_1 \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}), \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u})\big)_h, \tag{3.8}$$
$$T_7 := -2\nu\big(\alpha X(V), \tau_1 \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u})\big)_h, \tag{3.9}$$
$$T_8 := 2\nu\sigma\big(\tau_1 \boldsymbol{v}, \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u})\big)_h, \tag{3.10}$$
$$T_9 := 2\nu\big(\tau_1 \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}), \alpha X(U)\big)_h, \tag{3.11}$$
$$T_{10} := \big(\alpha X(V), \tau_1 \alpha X(U)\big)_h, \tag{3.12}$$
$$T_{11} := -\sigma\big(\tau_1 \alpha X(U), \boldsymbol{v}\big)_h, \tag{3.13}$$
$$T_{12} := 2\nu\sigma\big(\tau_1 \nabla\cdot(\alpha \overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}), \boldsymbol{u}\big)_h, \tag{3.14}$$
$$T_{13} := \sigma\big(\tau_1 \boldsymbol{u}, \alpha X(V)\big)_h, \tag{3.15}$$
$$T_{14} := -\sigma^2\big(\tau_1 \boldsymbol{u}, \boldsymbol{v}\big)_h, \tag{3.16}$$
$$T_{15} := -\varepsilon^2\big(\tau_2\, p, q\big)_h, \tag{3.17}$$
$$T_{16} := \varepsilon\big(\nabla\cdot(\alpha \boldsymbol{v}), \tau_2\, p\big)_h, \tag{3.18}$$
$$T_{17} := -\varepsilon\big(\tau_2 \nabla\cdot(\alpha \boldsymbol{u}), q\big)_h, \tag{3.19}$$
$$T_{18} := \big(\nabla\cdot(\alpha \boldsymbol{v}), \tau_2 \nabla\cdot(\alpha \boldsymbol{u})\big)_h. \tag{3.20}$$

Terms (3.3)–(3.7) come from the Galerkin part, (3.8)–(3.16) from the momentum stabilization and (3.17)–(3.20) from the mass stabilization in (1.4); they correspond, in this order, to terms (42)–(59) of [1].

Step 1: terms bounded directly by the working norm. By the Cauchy–Schwarz inequality (elementwise and then for the sums over elements),

$$|T_1| + |T_{10}| + |T_{18}| \le 2\nu \big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{u}\big\|\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla \boldsymbol{v}\big\| + \big\|\tau_1^{1/2}\alpha X(U)\big\|_h\big\|\tau_1^{1/2}\alpha X(V)\big\|_h + \big\|\tau_2^{1/2}\nabla\cdot(\alpha\boldsymbol{u})\big\|_h\big\|\tau_2^{1/2}\nabla\cdot(\alpha\boldsymbol{v})\big\|_h \;\le\; 2\,|||U|||\,|||V|||. \tag{3.21}$$

Step 2: the reactive pair. Since τ_1 is constant on each element and $\sigma - \sigma^2\tau_1 = \widetilde{\sigma}_\alpha$ on each element by (1.8),

$$T_3 + T_{14} = \big(\widetilde{\sigma}_\alpha\,\boldsymbol{u},\boldsymbol{v}\big)_h \le \big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{u}\big\|_h\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{v}\big\|_h \le |||U|||\,|||V|||. \tag{3.22}$$

Step 3: the penalty pair. By Lemma 2.1(P5), $\varepsilon\tau_{2,K} \le C_2 < 1$ on every element, hence

$$|T_5 + T_{15}| \le \varepsilon\,\|p\|\|q\| + \varepsilon\,C_2\,\|p\|\|q\| \le 2\,\varepsilon^{1/2}\|p\|\;\varepsilon^{1/2}\|q\| \le 2\,|||U|||\,|||V|||. \tag{3.23}$$

Step 4: terms containing the viscous operator. The key elemental estimate is obtained from (2.1) together with $\nu\tau_1\alpha_K \le h^2/c_1$ (Lemma 2.1(P2)): for any finite element velocity $\boldsymbol{w}_h$,

$$\big\|\tau_1^{1/2}\,2\nu\,\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{w}_h)\big\|_K \le 2\nu\,\tau_1^{1/2}\,\frac{\overline{C}_{\mathrm{inv}}}{h}\,\alpha_K^{1/2}\,\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{w}_h\big\|_K \le \frac{2\overline{C}_{\mathrm{inv}}}{c_1^{1/2}}\,\nu^{1/2}\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{w}_h\big\|_K. \tag{3.24}$$

Consequently, by the Cauchy–Schwarz inequality and (3.24) applied to $\boldsymbol{u}$ and/or $\boldsymbol{v}$,

$$\begin{aligned}|T_6| &\le \big\|\tau_1^{1/2}2\nu\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u})\big\|_h\big\|\tau_1^{1/2}2\nu\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{v})\big\|_h\\ &\le \frac{4\overline{C}_{\mathrm{inv}}^2}{c_1}\,\nu\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}\big\|\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{v}\big\| \le C\,|||U|||\,|||V|||,\end{aligned} \tag{3.25}$$

$$\begin{aligned}|T_7| &\le \big\|\tau_1^{1/2}\alpha X(V)\big\|_h\big\|\tau_1^{1/2}2\nu\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u})\big\|_h\\ &\le \frac{2\overline{C}_{\mathrm{inv}}}{c_1^{1/2}}\,\nu^{1/2}\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}\big\|\,\big\|\tau_1^{1/2}\alpha X(V)\big\|_h \le C\,|||U|||\,|||V|||,\end{aligned} \tag{3.26}$$

and $|T_9| \le C|||U|||\,|||V|||$ in the same way as T_7, with the roles of $\boldsymbol{u}$ and $\boldsymbol{v}$ interchanged. For T_8, the chained estimates (2.1)–(2.2) give the "double" inverse estimate

$$\big\|\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{w}_h)\big\|_K \le \frac{\overline{C}_{\mathrm{inv}}\,C_{\mathrm{inv}}}{h^2}\,\alpha_K\,\|\boldsymbol{w}_h\|_K, \tag{3.27}$$

whence, using $c_1\nu\alpha_K/h^2 \le \varphi_1$ (Lemma 2.1(P2)) and $\sigma\varphi_1\tau_1 = \widetilde{\sigma}_\alpha$,

$$\begin{aligned}|T_8| &\le \sum_K 2\nu\sigma\tau_1\,\frac{\overline{C}_{\mathrm{inv}}C_{\mathrm{inv}}}{h^2}\,\alpha_K\,\|\boldsymbol{u}\|_K\|\boldsymbol{v}\|_K = \frac{2\overline{C}_{\mathrm{inv}}C_{\mathrm{inv}}}{c_1}\sum_K\sigma\left(\frac{c_1\nu\alpha_K}{h^2}\right)\tau_1\,\|\boldsymbol{u}\|_K\|\boldsymbol{v}\|_K\\ &\le \frac{2\overline{C}_{\mathrm{inv}}C_{\mathrm{inv}}}{c_1}\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{u}\big\|_h\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{v}\big\|_h \le C\,|||U|||\,|||V|||,\end{aligned} \tag{3.28}$$

and $|T_{12}| \le C|||U|||\,|||V|||$ identically, applying (3.27) to $\boldsymbol{v}$ instead of $\boldsymbol{u}$. Summarizing this step,

$$|T_6| + |T_7| + |T_8| + |T_9| + |T_{12}| \le C\,|||U|||\,|||V|||. \tag{3.29}$$

Step 5: splitting of T_{13}. Write $T_{13} = \sigma(\tau_1\boldsymbol{u},\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{v})_h + \sigma(\tau_1\boldsymbol{u},\alpha\nabla q)_h$. For the first contribution, (2.4) and $c_2\alpha_K|\boldsymbol{a}|_\infty/h \le \varphi_1$ (Lemma 2.1(P2)) give

$$\begin{aligned}\big|\sigma(\tau_1\boldsymbol{u},\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{v})_h\big| &\le \sum_K \sigma\tau_1\,\alpha_K|\boldsymbol{a}|_\infty\frac{C_{\mathrm{inv}}}{h}\,\|\boldsymbol{u}\|_K\|\boldsymbol{v}\|_K \le \frac{C_{\mathrm{inv}}}{c_2}\sum_K\sigma\varphi_1\tau_1\|\boldsymbol{u}\|_K\|\boldsymbol{v}\|_K\\ &\le \frac{C_{\mathrm{inv}}}{c_2}\,\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{u}\big\|_h\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{v}\big\|_h.\end{aligned} \tag{3.30}$$

The second contribution, $R := \sigma(\tau_1 \boldsymbol{u}, \alpha\nabla q)_h$, is kept and treated in the next step.

Step 6: the convective–pressure group $N := R + T_2 + T_4 + T_{11}$.

Step 6a: exact rewriting of N. We first record three integration-by-parts identities.

(i) (Global skew-symmetry of convection.) For $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{V}_{h0}$ the pointwise identity $(\alpha\boldsymbol{a}\cdot\nabla\boldsymbol{u})\cdot\boldsymbol{v} + (\alpha\boldsymbol{a}\cdot\nabla\boldsymbol{v})\cdot\boldsymbol{u} = \alpha\boldsymbol{a}\cdot\nabla(\boldsymbol{u}\cdot\boldsymbol{v})$ holds, and $\boldsymbol{u}\cdot\boldsymbol{v} \in W_0^{1,1}(\Omega)$; Assumption 3 then gives $\int_\Omega \alpha\boldsymbol{a}\cdot\nabla(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Omega = 0$, i.e.

$$(\boldsymbol{v}, \alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{u}) = -(\boldsymbol{u}, \alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{v}). \tag{3.31}$$

(ii) (Global pressure/mass integration by parts.) Since $\alpha \in W^{1,\infty}(\Omega)$ and $\boldsymbol{v} \in H_0^1(\Omega)^d$, we have $\alpha\boldsymbol{v} \in H_0^1(\Omega)^d$, and similarly for $\boldsymbol{u}$; hence

$$(\boldsymbol{v}, \alpha\nabla p) = -\big(p, \nabla\cdot(\alpha\boldsymbol{v})\big), \qquad \big(q, \nabla\cdot(\alpha\boldsymbol{u})\big) = -(\boldsymbol{u}, \alpha\nabla q). \tag{3.32}$$

(iii) (Elementwise convective integration by parts.) On each K, with τ_1 constant on K, $\boldsymbol{a}$ continuous on $\overline{K}$ and $\nabla\cdot(\alpha\boldsymbol{a}) = 0$ in K, the divergence theorem gives

$$\big(\tau_1\,\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{u}, \boldsymbol{v}\big)_K = -\big(\tau_1\,\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{v}, \boldsymbol{u}\big)_K + \tau_1\int_{\partial K} \alpha\,(\boldsymbol{n}\cdot\boldsymbol{a})\,(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Gamma. \tag{3.33}$$

Using (3.31)–(3.32) in T_2 and T_4,

$$T_2 + T_4 = -\big(\boldsymbol{u}, \alpha X(V)\big) - \big(p, \nabla\cdot(\alpha\boldsymbol{v})\big),$$

while (3.33) applied to the convective part of T_{11} gives

$$T_{11} = \sigma\big(\tau_1\,\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{v}, \boldsymbol{u}\big)_h - \sigma\sum_K \tau_{1,K}\int_{\partial K}\alpha(\boldsymbol{n}_K\cdot\boldsymbol{a})(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Gamma - \sigma\big(\tau_1\boldsymbol{v}, \alpha\nabla p\big)_h.$$

Adding $R = \sigma(\tau_1\boldsymbol{u}, \alpha\nabla q)_h$ and collecting terms,

$$\begin{aligned} N = {} & \underbrace{-\sigma\big(\tau_1\boldsymbol{v}, \alpha\nabla p\big)_h}_{[\mathrm{I}]} + \underbrace{\sigma\big(\tau_1\boldsymbol{u}, \alpha X(V)\big)_h}_{[\mathrm{II}]} \underbrace{-\sigma\sum_K \tau_{1,K}\int_{\partial K}\alpha(\boldsymbol{n}_K\cdot\boldsymbol{a})(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Gamma}_{[\mathrm{III}]} \\ & \underbrace{-\big(p, \nabla\cdot(\alpha\boldsymbol{v})\big)}_{[\mathrm{IV}]} \underbrace{-\big(\boldsymbol{u}, \alpha X(V)\big)}_{[\mathrm{V}]}, \end{aligned} \tag{3.34}$$

which is the porous counterpart of Eq. (66) of [1]. (Note that the inter-element term [III] carries a *minus* sign; in [1] it is printed with a plus sign, a harmless typo since the term is estimated in absolute value, cf. Section 5.)

Step 6b: the pressure terms [I]+[IV]. Integrating by parts elementwise, $(\tau_1\boldsymbol{v}, \alpha\nabla p)_K = -(\tau_1\nabla\cdot(\alpha\boldsymbol{v}), p)_K + \tau_1\int_{\partial K}\alpha(\boldsymbol{n}\cdot\boldsymbol{v})\,p\,\mathrm{d}\Gamma$, and using $1 - \sigma\tau_1 = \varphi_1\tau_1$ on each element,

$$-\big([\mathrm{I}] + [\mathrm{IV}]\big) = \sigma\big(\tau_1\boldsymbol{v}, \alpha\nabla p\big)_h + \big(p, \nabla\cdot(\alpha\boldsymbol{v})\big) = \big(\varphi_1\tau_1\,p, \nabla\cdot(\alpha\boldsymbol{v})\big)_h + \sigma\sum_b\int_{\Gamma^b} [\![\tau_1]\!]\,\alpha\,(\boldsymbol{n}\cdot\boldsymbol{v})\,p\,\mathrm{d}\Gamma, \tag{3.35}$$

where the sum extends over the interior faces Γ^b of the partition: the elemental boundary contributions have been assembled facewise using the continuity of $\alpha, \boldsymbol{v}, p$ across interior faces ($\boldsymbol{n}$ is a fixed unit normal on each face and $[\![\tau_1]\!]$ the corresponding jump), while the faces lying on Γ vanish because $\boldsymbol{v} = \boldsymbol{0}$ there.

For the volumetric part of (3.35), on each element $\varphi_1\tau_1 \le \varphi_1^{1/2}\tau_1^{1/2}$ (since $\varphi_1\tau_1 \le 1$, Lemma 2.1(P1)), and $\varphi_1^{1/2} = c_1^{1/2}\alpha_K\tau_2^{1/2}/h$ (Lemma 2.1(P3)), so that

$$\big|(\varphi_1\tau_1\, p, \nabla\cdot(\alpha\boldsymbol{v}))_h\big| \le c_1^{1/2}\sum_K \tau_2^{1/2}\big\|\nabla\cdot(\alpha\boldsymbol{v})\big\|_K\, \alpha_K\, \tau_1^{1/2}h^{-1}\|p\|_K \le c_1^{1/2}\, |||V|||\, \big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h. \tag{3.36}$$

For the jump part, fix an interior face Γ^b shared by K_1 and K_2 and set $\Omega^b = K_1\cup K_2$. By Assumption 5 the elemental quantities $\varphi_1, \tau_1, \widetilde{\sigma}_\alpha$ of K_1 and K_2 are uniformly comparable, so Lemma 2.4 yields, for any $i,j\in\{1,2\}$,

$$\sigma\big|[\![\tau_1]\!]\big| \le C\left(\widetilde{\sigma}_\alpha\tau_1\big|_{K_i}\right)^{1/2}\left(\widetilde{\sigma}_\alpha\tau_1\big|_{K_j}\right)^{1/2} \le C\,\widetilde{\sigma}_\alpha^{1/2}\big|_{K_i}\,\tau_{1,K_j}^{1/2}, \tag{3.37}$$

the last step using $\widetilde{\sigma}_\alpha\tau_1 \le 1$ (Lemma 2.1(P4)) on K_j together with the comparability of τ_{1,K_i} and τ_{1,K_j}. Moreover $\alpha \le \alpha_{K_j}$ on $\Gamma^b \subset \overline{K}_j$. Bounding the face integral by the L^∞ norms over Ω^b, expanding $\|\boldsymbol{v}\|_{L^\infty(\Omega^b)}\|p\|_{L^\infty(\Omega^b)} \le \sum_{i,j\in\{1,2\}}\|\boldsymbol{v}\|_{L^\infty(K_i)}\|p\|_{L^\infty(K_j)}$, and applying the L^∞ inverse estimate (1.12) on each element (the diameters of K_1, K_2 being comparable by Assumption 4), together with $\operatorname{meas}(\Gamma^b) \le Ch^{d-1}$,

$$\sigma\left|\int_{\Gamma^b}[\![\tau_1]\!]\alpha(\boldsymbol{n}\cdot\boldsymbol{v})p\,\mathrm{d}\Gamma\right| \le C\sum_{i,j\in\{1,2\}}\widetilde{\sigma}_\alpha^{1/2}\big|_{K_i}\|\boldsymbol{v}\|_{K_i}\,\alpha_{K_j}\,\tau_{1,K_j}^{1/2}\,h^{-1}\|p\|_{K_j}. \tag{3.38}$$

Summing over the interior faces and applying the Cauchy–Schwarz inequality to the sums (each element meets a uniformly bounded number of faces, Assumption 4),

$$\sigma\sum_b\left|\int_{\Gamma^b}[\![\tau_1]\!]\alpha(\boldsymbol{n}\cdot\boldsymbol{v})p\,\mathrm{d}\Gamma\right| \le C\,\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{v}\big\|_h\,\big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h \le C\,|||V|||\,\big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h. \tag{3.39}$$

Combining (3.36) and (3.39),

$$\big|[\mathrm{I}]+[\mathrm{IV}]\big| \le C\,|||V|||\,\big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h. \tag{3.40}$$

Step 6c: the terms [II]+[V]. Again with $1-\sigma\tau_1 = \varphi_1\tau_1$ elementwise, and using $\varphi_1^{1/2}\tau_1 \le \tau_1^{1/2}$ together with Lemma 2.1(P3),

$$\begin{aligned}\big|[\mathrm{II}]+[\mathrm{V}]\big| = \big|(\varphi_1\tau_1\,\boldsymbol{u}, \alpha X(V))_h\big| &\le \sum_K \varphi_1^{1/2}\|\boldsymbol{u}\|_K\,\varphi_1^{1/2}\tau_1\big\|\alpha X(V)\big\|_K\\ &\le c_1^{1/2}\sum_K \alpha_K\tau_2^{1/2}h^{-1}\|\boldsymbol{u}\|_K\,\tau_1^{1/2}\big\|\alpha X(V)\big\|_K\\ &\le c_1^{1/2}\,\big\|\alpha_K\tau_2^{1/2}h^{-1}\boldsymbol{u}\big\|_h\,|||V|||.\end{aligned} \tag{3.41}$$

Step 6d: the inter-element term [III]. As in Step 6b, the elemental boundary contributions assemble facewise (the integrand $\alpha(\boldsymbol{n}\cdot\boldsymbol{a})(\boldsymbol{u}\cdot\boldsymbol{v})$ is single-valued on interior faces and $\boldsymbol{u}$ vanishes on Γ):

$$[\mathrm{III}] = -\sigma\sum_b\int_{\Gamma^b}[\![\tau_1]\!]\,\alpha(\boldsymbol{n}\cdot\boldsymbol{a})(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Gamma.$$

On Γ^b we bound $\alpha\,|\boldsymbol{n}\cdot\boldsymbol{a}| \le \alpha_{K_j}|\boldsymbol{a}|_{\infty,K_j} \le \varphi_{1,K_j}h_{K_j}/c_2$ (Lemma 2.1(P2)). Using (3.37) in the symmetric form $\sigma|[\![\tau_1]\!]| \le C(\widetilde{\sigma}_\alpha\tau_1|_{K_i})^{1/2}(\widetilde{\sigma}_\alpha\tau_1|_{K_j})^{1/2}$, and $\tau_{1,K_i}^{1/2}\varphi_{1,K_j}^{1/2} \le C\,(\tau_1\varphi_1|_{K_j})^{1/2} \le C$ (comparability of τ_1 across the face and $\varphi_1\tau_1 \le 1$), we obtain on each face, after applying the L^∞ inverse estimates and $\operatorname{meas}(\Gamma^b) \le Ch^{d-1}$ exactly as in Step 6b,

$$\sigma\left|\int_{\Gamma^b}[\![\tau_1]\!]\alpha(\boldsymbol{n}\cdot\boldsymbol{a})(\boldsymbol{u}\cdot\boldsymbol{v})\,\mathrm{d}\Gamma\right| \le C\sum_{i,j\in\{1,2\}}\widetilde{\sigma}_\alpha^{1/2}\big|_{K_i}\|\boldsymbol{u}\|_{K_i}\,\widetilde{\sigma}_\alpha^{1/2}\big|_{K_j}\|\boldsymbol{v}\|_{K_j}, \tag{3.42}$$

whence, summing over faces and applying the Cauchy–Schwarz inequality as before,

$$\big|[\mathrm{III}]\big| \le C\,\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{u}\big\|_h\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{v}\big\|_h \le C\,|||U|||\,|||V|||. \tag{3.43}$$

Collecting (3.30), (3.40), (3.41) and (3.43),

$$|T_{13}| + |N| \le C\,|||V|||\Big(|||U||| + \big\|\alpha_K\tau_2^{1/2}h^{-1}\boldsymbol{u}\big\|_h + \big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h\Big). \tag{3.44}$$

Step 7: the penalty cross terms. By Lemma 2.1(P5), on each element $\varepsilon\tau_2^{1/2} = (\varepsilon\tau_2)^{1/2}\varepsilon^{1/2} \le C_2^{1/2}\varepsilon^{1/2}$, hence

$$|T_{16}| \le \big\|\tau_2^{1/2}\nabla\cdot(\alpha\boldsymbol{v})\big\|_h\,\varepsilon\big\|\tau_2^{1/2}p\big\|_h \le C_2^{1/2}\,|||V|||\,\varepsilon^{1/2}\|p\| \le |||V|||\,|||U|||, \tag{3.45}$$

and likewise $|T_{17}| \le |||U|||\,|||V|||$.

Step 8: assembly. Adding the estimates of Steps 1–7,

$$B_{\mathrm{ASGS}}(\boldsymbol{a};U,V) \le C\,|||V|||\Big(|||U||| + \big\|\alpha_K\tau_2^{1/2}h^{-1}\boldsymbol{u}\big\|_h + \big\|\alpha_K\tau_1^{1/2}h^{-1}p\big\|_h\Big). \tag{3.46}$$

Step 9: absorption of the working norm. It remains to bound $|||U|||$ for *discrete* $U = U_h$ by the bracketed norms. We estimate each contribution to (1.9) elementwise, using Lemmas 2.1 and 2.2:

$$\nu^{1/2}\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}\big\|_K \le \nu^{1/2}\frac{C_{\mathrm{inv}}}{h}\alpha_K^{1/2}\|\boldsymbol{u}\|_K \le \frac{C_{\mathrm{inv}}}{c_1^{1/2}}\,\varphi_1^{1/2}\|\boldsymbol{u}\|_K = C_{\mathrm{inv}}\,\alpha_K\,\tau_2^{1/2}h^{-1}\|\boldsymbol{u}\|_K, \tag{3.47}$$

$$\big\|\widetilde{\sigma}_\alpha^{1/2}\boldsymbol{u}\big\|_K \le \varphi_1^{1/2}\|\boldsymbol{u}\|_K = c_1^{1/2}\,\alpha_K\,\tau_2^{1/2}h^{-1}\|\boldsymbol{u}\|_K, \tag{3.48}$$

$$\varepsilon^{1/2}\|p\|_K \le c_1^{1/2}\,\alpha_K\,\tau_1^{1/2}h^{-1}\|p\|_K, \tag{3.49}$$

$$\begin{aligned}\tau_1^{1/2}\big\|\alpha X(U)\big\|_K &\le \tau_1^{1/2}\alpha_K|\boldsymbol{a}|_\infty\frac{C_{\mathrm{inv}}}{h}\|\boldsymbol{u}\|_K + \tau_1^{1/2}\alpha_K\frac{C_{\mathrm{inv}}}{h}\|p\|_K \\ &\le \frac{C_{\mathrm{inv}}\,c_1^{1/2}}{c_2}\,\alpha_K\tau_2^{1/2}h^{-1}\|\boldsymbol{u}\|_K + C_{\mathrm{inv}}\,\alpha_K\,\tau_1^{1/2}h^{-1}\|p\|_K,\end{aligned} \tag{3.50}$$

$$\tau_2^{1/2}\big\|\nabla\cdot(\alpha\boldsymbol{u})\big\|_K \le \overline{C}_{\mathrm{inv}}\,\alpha_K\,\tau_2^{1/2}h^{-1}\|\boldsymbol{u}\|_K. \tag{3.51}$$

Here (3.47) uses (2.2) and (P2)–(P3); (3.48) uses (P4) and (P3); (3.49) uses (P5); in (3.50) the velocity part uses (2.4), $\tau_1^{1/2}\alpha_K|\boldsymbol{a}|_\infty/h \le \varphi_1\tau_1^{1/2}/c_2 \le \varphi_1^{1/2}/c_2$ and (P3), while the pressure part uses (2.4) directly; and (3.51) uses (2.3). Squaring and summing over the elements,

$$|||U_h||| \le C\Big(\big\|\alpha_K\tau_2^{1/2}h^{-1}\boldsymbol{u}_h\big\|_h + \big\|\alpha_K\tau_1^{1/2}h^{-1}p_h\big\|_h\Big). \tag{3.52}$$

Inserting (3.52) in (3.46) proves (3.2), and $\alpha_K \le 1$ (Assumption 2) gives (3.1). □

Remark 3.2 (Sharper, porosity-weighted estimate). The bound (3.2) is strictly sharper than (3.1) wherever α is small, and it propagates to the convergence estimate (cf. Remark 4.4). It is remarkable that the smallness condition (1.10) on ε, with its factor α_K^2, is exactly what is needed for the penalty contribution (3.49) to carry the weight α_K as well; the same factor is the one required by the stability analysis of the main text.

Remark 3.3 (Where each hypothesis is used). Assumption 1 (condition (1.10)) enters Steps 3, 7 and 9; Assumption 2 enters through Lemma 2.2 (Steps 4, 5 and 9) and through $\alpha_K \le 1$; Assumption 3 enters Step 6a; Assumption 4 enters every inverse estimate and the facewise bookkeeping of Steps 6b and 6d; Assumption 5 enters only (3.37), i.e. the jump terms of Steps 6b and 6d, which carry the factor σ and hence vanish identically when $\sigma = 0$. As in [1], the jump condition and the L^∞ inverse estimate are therefore needed only when $\sigma > 0$.

4 Consequences for the convergence analysis

4.1 Interpolation estimates

Let $U = [\boldsymbol{u}; p]$ denote the solution of the continuous linearized problem and assume the regularity

$$U \in \mathcal{W}_r := \big(H^m(\Omega)\cap H_0^1(\Omega)\big)^d \times \big(H^n(\Omega)\cap \mathcal{Q}_0\big), \qquad m \geq k_u+1\ (\geq 2), \quad n \geq k_p+1\ (\geq 2), \tag{4.1}$$

(see Section 5 concerning the exponent n). In particular $\mathcal{L}_{\boldsymbol{a}} U \in L^2(K)$ for every K, so that the consistency property (1.5) holds, and for $d \leq 3$ both $\boldsymbol{u}$ and p are continuous on $\overline{\Omega}$. Let $\widehat{\boldsymbol{u}}_h \in \mathcal{V}_{h0}$ and $\widehat{p}_h \in \mathcal{Q}_{h0}$ be interpolants of $\boldsymbol{u}$ and p (e.g. the Lagrange interpolants), set $\widehat{U}_h := [\widehat{\boldsymbol{u}}_h; \widehat{p}_h]$, and define the elemental interpolation errors

$$\mathrm{E}_{\mathrm{int}}^K(\boldsymbol{u}) := h_K^{k_u+1} \|\boldsymbol{u}\|_{H^{k_u+1}(K)}, \qquad \mathrm{E}_{\mathrm{int}}^K(p) := h_K^{k_p+1} \|p\|_{H^{k_p+1}(K)}. \tag{4.2}$$

Standard interpolation theory [2, 3] provides, on every K,

$$\begin{aligned} \|\boldsymbol{u} - \widehat{\boldsymbol{u}}_h\|_{H^{m'}(K)} &\leq C\, h_K^{-m'}\, \mathrm{E}_{\mathrm{int}}^K(\boldsymbol{u}), \qquad m' \in \{0,1,2\}, \\ \|p - \widehat{p}_h\|_{H^{m'}(K)} &\leq C\, h_K^{-m'}\, \mathrm{E}_{\mathrm{int}}^K(p), \qquad m' \in \{0,1\}, \end{aligned} \tag{4.3}$$

and, since $k_u + 1,\ k_p + 1 > d/2$ for $d \leq 3$, also the L^∞ estimates

$$\|\boldsymbol{u} - \widehat{\boldsymbol{u}}_h\|_{L^\infty(K)} \leq C\, h_K^{-d/2}\, \mathrm{E}_{\mathrm{int}}^K(\boldsymbol{u}), \qquad \|p - \widehat{p}_h\|_{L^\infty(K)} \leq C\, h_K^{-d/2}\, \mathrm{E}_{\mathrm{int}}^K(p), \tag{4.4}$$

which are the analogues of Eqs. (78)–(79) of [1]. We abbreviate $E := U - \widehat{U}_h = [\boldsymbol{e}_u; e_p]$ and define

$$\psi(h) := \sum_{K\in\mathcal{T}_h} \alpha_K\, h_K^{-1} \Big(\tau_{2,K}^{1/2}\, \mathrm{E}_{\mathrm{int}}^K(\boldsymbol{u}) + \tau_{1,K}^{1/2}\, \mathrm{E}_{\mathrm{int}}^K(p)\Big) \ \leq \ \sum_{K\in\mathcal{T}_h} h_K^{-1} \Big(\tau_{2,K}^{1/2}\, \mathrm{E}_{\mathrm{int}}^K(\boldsymbol{u}) + \tau_{1,K}^{1/2}\, \mathrm{E}_{\mathrm{int}}^K(p)\Big). \tag{4.5}$$

4.2 Continuity with respect to the interpolation error

Lemma 4.1 (Continuity for the interpolation error)**.** *Let Assumptions 1 to 5 hold and let $U \in \mathcal{W}_r$. Then*

$$B_{\mathrm{ASGS}}\big(\boldsymbol{a};\, U - \widehat{U}_h,\, V_h\big) \leq C\, \psi(h)\, |||V_h||| \qquad \forall\, V_h \in \mathcal{X}_{h0}. \tag{4.6}$$

Proof. We rerun the proof of Lemma 3.1 with first argument $E = [\boldsymbol{e}_u; e_p]$ and second argument V_h, which remains discrete and is treated exactly as before. Two observations make this possible.

(1) The exact identities remain valid. The decomposition of Step 0 and the identities of Steps 2, 3 and 6a–6b only require of the first argument that $\boldsymbol{e}_u \in H_0^1(\Omega)^d$ with $\boldsymbol{e}_u|_K \in H^2(K)^d$, and $e_p \in H^1(\Omega)$, together with the continuity of $\boldsymbol{e}_u$ and e_p across interior faces; all of this holds by (4.1) (for $d \leq 3$, $H^2(K) \hookrightarrow C^0(\overline{K})$ and $\boldsymbol{u}, p \in C^0(\overline{\Omega})$, while $\widehat{\boldsymbol{u}}_h, \widehat{p}_h$ are continuous finite element functions). In particular $\boldsymbol{e}_u \cdot \boldsymbol{v}_h \in W_0^{1,1}(\Omega)$, so (3.31) holds, and the elementwise identities (3.33) and those preceding (3.35) hold with well-defined traces.

(2) Inverse estimates on the first argument are replaced by interpolation estimates. Inspection of the proof of Lemma 3.1 shows that the first argument enters the estimates only through the following elemental quantities; next to each we record the discrete estimate used there and its

replacement, obtained from (1.11), (4.3)–(4.4) and $|\overset{\mathrm{DS}}{\Pi}\mathbf{T}| \le |\mathbf{T}|$:

$$\big\|\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{e}_u)\big\|_K \le \alpha_{\infty,K}\big\|\nabla\cdot(\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{e}_u)\big\|_K + \|\nabla\alpha\|_{L^\infty(K)}\big\|\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{e}_u\big\|_K \le C\,\alpha_K\,h^{-2}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \tag{4.7}$$

$$\big\|\alpha^{1/2}\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{e}_u\big\|_K \le \alpha_K^{1/2}\,|\boldsymbol{e}_u|_{H^1(K)} \le C\,\alpha_K^{1/2}\,h^{-1}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \tag{4.8}$$

$$\big\|\nabla\cdot(\alpha\boldsymbol{e}_u)\big\|_K \le C\,\alpha_K\,h^{-1}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \qquad \|\alpha\,\boldsymbol{a}\cdot\nabla\boldsymbol{e}_u\|_K \le C\,\alpha_K\,|\boldsymbol{a}|_{\infty,K}\,h^{-1}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \tag{4.9}$$

$$\|\alpha\nabla e_p\|_K \le C\,\alpha_K\,h^{-1}\,\mathrm{E}^K_{\mathrm{int}}(p), \qquad \|\boldsymbol{e}_u\|_K \le C\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \qquad \|e_p\|_K \le C\,\mathrm{E}^K_{\mathrm{int}}(p), \tag{4.10}$$

$$\|\boldsymbol{e}_u\|_{L^\infty(K)} \le C\,h^{-d/2}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}), \qquad \|e_p\|_{L^\infty(K)} \le C\,h^{-d/2}\,\mathrm{E}^K_{\mathrm{int}}(p). \tag{4.11}$$

These have exactly the same elemental coefficients as the weighted inverse estimates of Lemma 2.2 applied in Steps 4, 5, 6 and 9, with $\|\boldsymbol{u}_h\|_K$ and $\|p_h\|_K$ replaced by $\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u})$ and $\mathrm{E}^K_{\mathrm{int}}(p)$. Consequently:

- Step 9 (the elemental bounds (3.47)–(3.51)) carried out with (4.7)–(4.11) gives

$$|||E||| \le C\Big(\sum_K \alpha_K^2\,h_K^{-2}\big(\tau_{2,K}(\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}))^2 + \tau_{1,K}(\mathrm{E}^K_{\mathrm{int}}(p))^2\big)\Big)^{1/2} \le C\,\psi(h), \tag{4.12}$$

  the last step by the discrete $\ell^2 \subset \ell^1$ inequality.

- The bracketed norms in (3.46) are bounded, by the $m' = 0$ estimates in (4.10), as $\big\|\alpha_K\tau_2^{1/2}h^{-1}\boldsymbol{e}_u\big\|_h + \big\|\alpha_K\tau_1^{1/2}h^{-1}e_p\big\|_h \le C\,\psi(h)$.

- In the jump terms (3.38) and (3.42) the L^∞ inverse estimates applied to the *first* argument are replaced by the L^∞ interpolation estimates (4.11) (those applied to $\boldsymbol{v}_h$ are kept), with identical powers of h.

Steps 1–8 then yield $B_{\mathrm{ASGS}}(\boldsymbol{a};E,V_h) \le C|||V_h|||\big(|||E||| + C\psi(h)\big)$, and (4.12) concludes the proof. □

4.3 Convergence

For completeness we recall the stability result of the main text, whose proof is given there (with the amendments of Remark 2.3 and Section 5).

Proposition 4.2 (Stability; main text). *Let Assumptions 1 to 4 hold and let $c_1 > 2\xi\,\overline{C}_{\mathrm{inv}}^2$ for some $\xi > 2$. Then there is $C > 0$ such that $B_{\mathrm{ASGS}}(\boldsymbol{a};U_h,U_h) \ge C\,|||U_h|||^2$ for all $U_h \in \mathcal{X}_{h0}$.*

Theorem 4.3 (Convergence). *Under the hypotheses of Proposition 4.2 and Lemma 4.1,*

$$|||U - U_h||| \le C\,\psi(h) \le C\sum_{K\in\mathcal{T}_h} h_K^{-1}\Big(\tau_{2,K}^{1/2}\,\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}) + \tau_{1,K}^{1/2}\,\mathrm{E}^K_{\mathrm{int}}(p)\Big). \tag{4.13}$$

Proof. Let $E_h := \widehat{U}_h - U_h \in \mathcal{X}_{h0}$. By Proposition 4.2, consistency (1.5) and Lemma 4.1,

$$C\,|||E_h|||^2 \le B_{\mathrm{ASGS}}(\boldsymbol{a};E_h,E_h) = B_{\mathrm{ASGS}}\big(\boldsymbol{a};\widehat{U}_h - U, E_h\big) \le C\,\psi(h)\,|||E_h|||,$$

so $|||E_h||| \le C\psi(h)$. The triangle inequality and (4.12) give $|||U - U_h||| \le |||U - \widehat{U}_h||| + |||E_h||| \le C\psi(h)$. □

Remark 4.4 (Sharper convergence estimate). Since the proof goes through with the porosity-weighted quantity $\psi(h)$ of (4.5), the convergence estimate of the main text in fact holds in the slightly sharper form $|||U - U_h||| \le C\sum_K \alpha_K h_K^{-1}(\tau_{2,K}^{1/2}\mathrm{E}^K_{\mathrm{int}}(\boldsymbol{u}) + \tau_{1,K}^{1/2}\mathrm{E}^K_{\mathrm{int}}(p))$, with the local weight $\alpha_K \le 1$. This refinement is consistent with (and slightly strengthens) the robustness discussion of the main text in regions of low porosity.

5 Amendments to the original statements

We collect here the small corrections that the above analysis introduces with respect to the statements in the main text and in [1].

(A1) *Smallness condition on ε (main text).* The condition should be stated in the two-constant form (1.10), $\varepsilon \le C_2\, c_1 \min_K\{\alpha_K^2 \tau_{1,K}/h_K^2\}$ with a fixed $C_2 < 1$, rather than as a strict inequality with $C_2 = 1$. Indeed, with the latter one has $\varepsilon\tau_{2,K} < \varphi_{1,K}\tau_{1,K} \le 1$ but *without a uniform margin* when $\sigma = 0$, so the coercivity of the penalty block, $\varepsilon(1-\varepsilon\tau_2)\|p_h\|^2 \ge C\varepsilon\|p_h\|^2$, would not hold with a constant independent of the data. The same uniform margin ($\varepsilon\tau_2 \le C_2 < 1$) is used here in Steps 3 and 7, and the factor α_K^2 in (1.10) is what allows the penalty term to be absorbed with the weight α_K in (3.49). This mirrors Eq. (34) of [1].

(A2) *Definition of the broken norm (main text).* The elementwise norm must be defined as the ℓ^2 combination (1.3), $\|\tau^{1/2} f\|_h = (\sum_K \tau_K \|f\|_K^2)^{1/2}$, and not as the plain sum $\sum_K \|f\|_{L^2(K)}$; otherwise the working norm (1.9) and all Cauchy–Schwarz manipulations would be inconsistent.

(A3) *Hypotheses of the continuity lemma (main text).* Besides positivity of the algorithmic constants and $\nabla\cdot(\alpha\boldsymbol{a}) = 0$, the proof requires: the porosity resolution condition (1.11) (assumed earlier in the analysis section of the main text, but to be included among the hypotheses of the lemma); the smallness condition (1.10); when $\sigma > 0$, the jump condition (1.13) together with the L^∞ inverse estimate in (1.12) (the exact analogue of conditions (38)–(39) of [1]); and trace regularity of $\boldsymbol{a}$ (Assumption 3). The trial and test pairs should range over $\mathcal{X}_{h0} = \mathcal{X}_{h0}$ (homogeneous boundary values), since the boundary faces must drop from the inter-element terms.

(A4) *Inverse estimates for α-weighted quantities (main text).* Since α is not piecewise polynomial, the inverse estimates applied to $\nabla\cdot(\alpha\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}_h)$ and $\nabla\cdot(\alpha\boldsymbol{u}_h)$ hold with the constant $\overline{C}_{\mathrm{inv}} = \sqrt{d\,\delta_\alpha}\, C_{\mathrm{inv}} + C_\alpha$ of Lemma 2.2 rather than with the bare C_{inv}; accordingly, the lower bounds imposed on c_1 in the stability lemma should be read with $\overline{C}_{\mathrm{inv}}$ in place of C_{inv} (cf. Remark 2.3).

(A5) *Sign of the inter-element term in Eq. (66) of [1].* The elementwise integration by parts of the convective stabilization term produces the inter-element contribution with a *negative* sign, cf. (3.34); in [1] it is printed with a positive sign. The slip is harmless: the term is subsequently estimated in absolute value.

(A6) *The term $\frac{1}{2}(\nabla\cdot\boldsymbol{a})\boldsymbol{v}_h$ in [1].* In the original proof (Eqs. (65) and (74)) the contribution of $\frac{1}{2}(\nabla\cdot\boldsymbol{a})\boldsymbol{v}_h$ to $X(\boldsymbol{v}_h, q_h)$ is absorbed without comment; estimating it as the convective part requires control of $\|\nabla\cdot\boldsymbol{a}\|_{L^\infty(K)}$ by $|\boldsymbol{a}|_{\infty,K}/h_K$, which holds, e.g., when $\boldsymbol{a}$ is a finite element function (by an inverse estimate), but not for general $\boldsymbol{a} \in L^\infty$. In the porous formulation analyzed here the issue does not arise: under $\nabla\cdot(\alpha\boldsymbol{a}) = 0$ the convective operator is $\alpha\boldsymbol{a}\cdot\nabla$ with no skew term.

(A7) *Pressure regularity in Eq. (77) of [1].* The regularity space is stated with $n \ge k_p$; since the interpolation error $E_{\mathrm{int}}(p)$ involves $\|p\|_{H^{k_p+1}}$, it should read $n \ge k_p + 1$, as in (4.1).

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